Maximum Power Point Tracking of a Photovoltaic Module

ELEC5206 Sustainable Energy Systems Laboratory Report

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Abstract

A perturb-and-observe hill-climbing controller was implemented on a PICAXE 14M2 to adjust the duty cycle of a buck converter connected between a photovoltaic module and a 2 ohm resistive load. Two approximately 200 s records were analysed. Under the initial panel orientation, mean PV power was 4.064 W and the recorded peak was 4.463 W. After the PV panel was rotated by 60 degrees from its initial position, mean power fell to 2.153 W and the peak was 2.336 W. In both records, voltage and power repeatedly moved between nearby levels rather than diverging, which is the expected steady-state behaviour of fixed-step hill climbing. The controller therefore maintained operation in the neighbourhood of the available maximum, although the data do not include a simultaneous independent I-V sweep and cannot quantify absolute tracking efficiency.

1 Introduction and theory

The experiment aimed to implement maximum power point tracking (MPPT), verify that the controller settles near the available maximum power point (MPP), and assess its response when the PV panel was rotated by 60 degrees from its initial orientation. For a fixed resistor, output power is

$$P_o = V_o^2 / R_l$$

Because R_l is constant and V_o is non-negative, maximising load voltage is exactly equivalent to maximising load power. The controller can therefore use only the sensed output voltage for its hill-climbing decision. This removes the current sensor from the real-time control loop. It does not mean that current is physically constant; it means that voltage alone is a sufficient monotonic proxy for output power under a fixed resistive load.

2 Methodology and procedure

The PV module was illuminated by two 500 W halogen lamps and connected to a 2 ohm load through the laboratory buck converter. A 12 V supply powered the gate-driver circuit and a 5 V supply powered the PICAXE 14M2 controller. The DACI 9063 and LVDAC-EMS recorded PV voltage, current and power at 1 s intervals. An oscilloscope was used to observe PWM, PV voltage and converter output voltage. The connections were checked by a tutor before the lamps were energised.

Step	Procedure and control purpose
1	With lamps off, connect PV positive to Vin, load to Vo, and join PV/load negatives at Gnd.
2	Connect 12 V driver and 5 V controller supplies with common ground; connect PWM and voltage-sense leads.
3	Program the hill-climbing routine with D = 0.10, perturbation step ΔD = 0.01 and previous voltage V_old = 0.
4	After tutor approval, energise the lamps, allow thermal settling, and record 200 nominal one-second samples.
5	Rotate the PV panel by 60 degrees from its initial position, repeat the test, and export voltage, current and power.

3 Results

The supplied CSV files each contain 201 valid records. Because a few timestamps are skipped, the observed interval is 0-203 s for Condition 1 and 0-202 s for Condition 2. All calculations below use the recorded values without smoothing.

Test	Mean V (V)	Mean I (A)	Mean P (W)	P SD (W)	Min P (W)	Max P (W)
Initial orientation	9.440	0.430	4.064	0.153	3.752	4.463
Panel rotated 60 degrees	6.770	0.318	2.153	0.094	1.956	2.336

Table 1 Summary of measured PV quantities

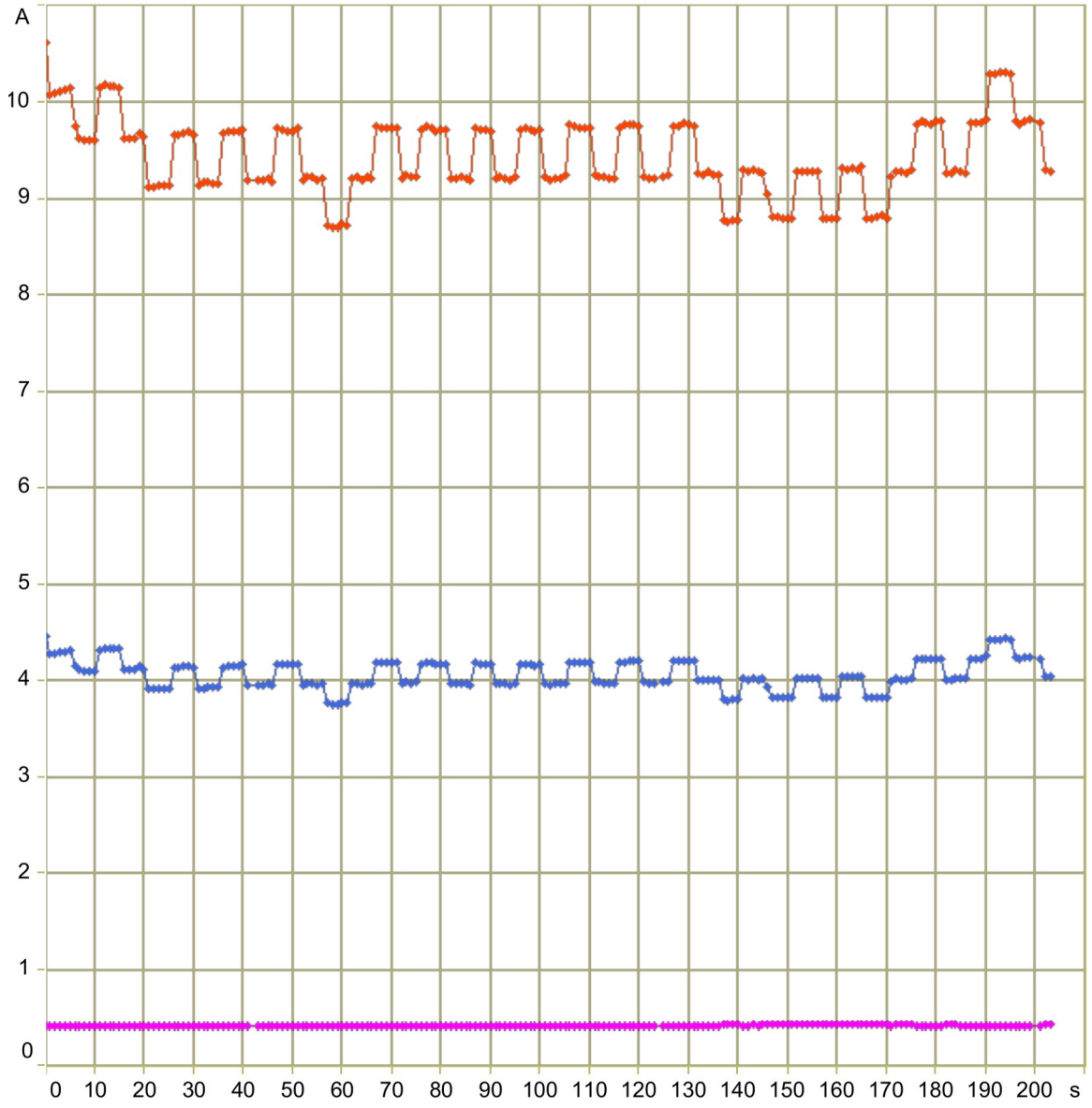


Figure 1 Original LVDAC-EMS trace at the initial panel orientation red voltage blue power magenta current

Condition 1 begins at 4.463 W and then exhibits repeated small power changes. The 4.064 W overall mean and 0.153 W standard deviation reflect the fixed-step search and the slow drift of the illuminated module. The last 20 samples average 4.217 W, showing that useful power remained available at the end of the test.

3 Results continued

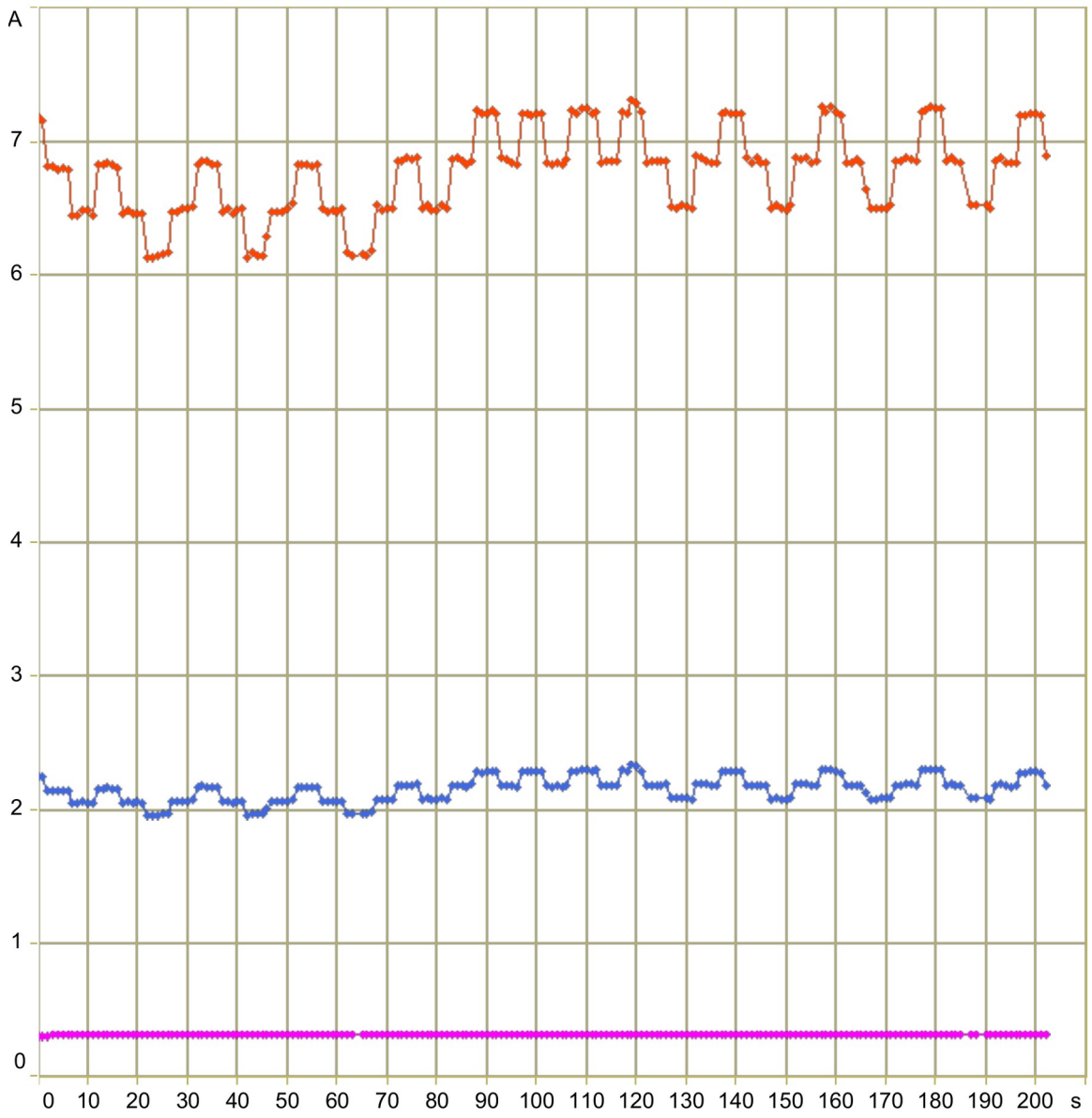


Figure 2 Original LVDAC-EMS trace after 60 degree panel rotation red voltage blue power magenta current

After the panel was rotated by 60 degrees from its initial position, the system produced a lower operating level: mean voltage fell from 9.440 V to 6.770 V and mean power from 4.064 W to 2.153 W. The 47.0% reduction in mean power is consistent with the lower effective irradiance expected when the panel surface is less directly aligned with the lamps. Despite this reduction, the controller continued to move among neighbouring operating points. The peak of 2.336 W occurred at 119 s, while the final 20-s mean was 2.181 W.

3.1 Unit labelled waveform comparison

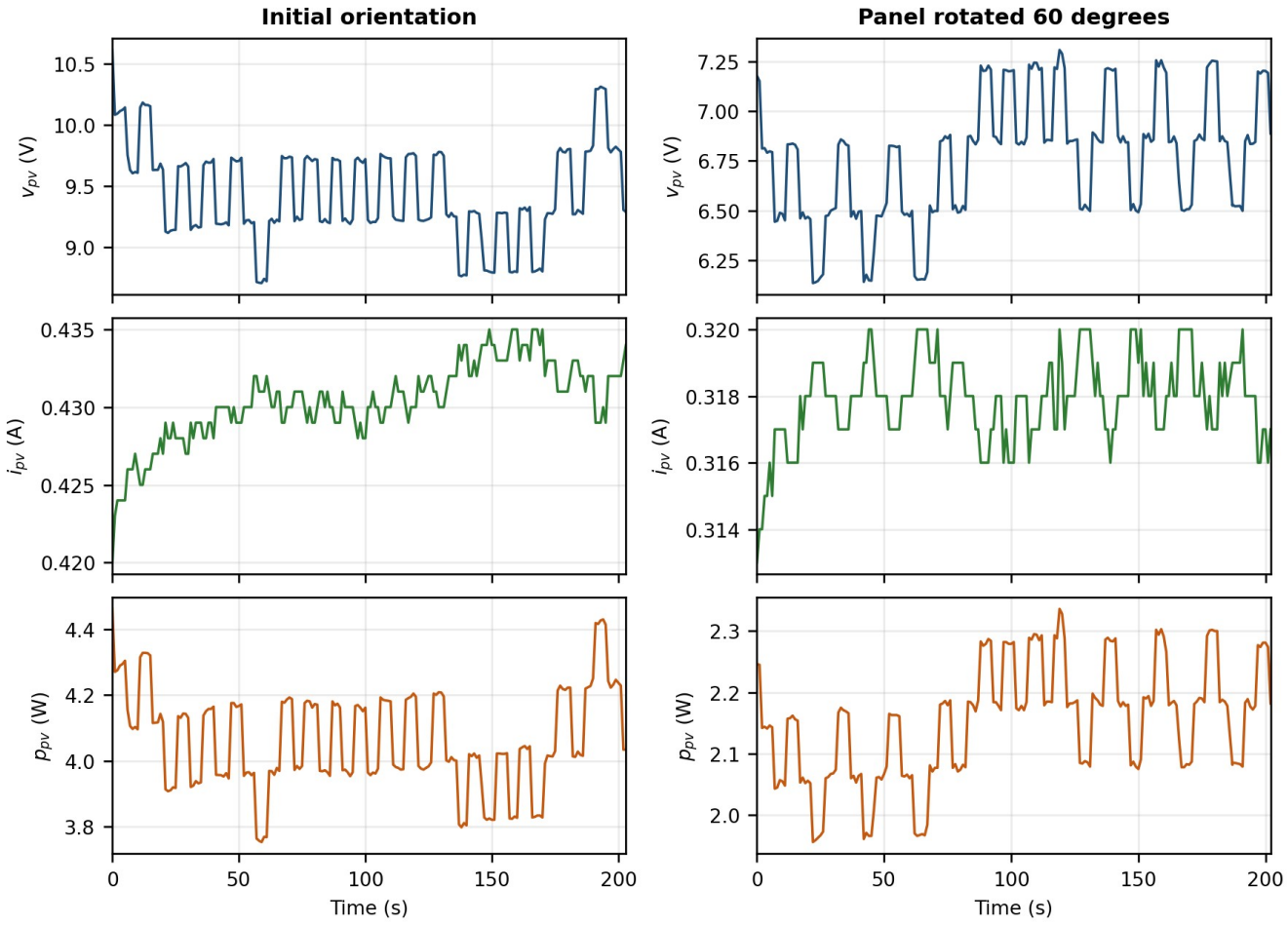


Figure 3 Separate voltage current and power waveforms derived from the complete CSV records

Condition	Time (s)	Voltage (V)	Current (A)	Power (W)
Initial	0	10.610	0.420	4.463
Initial	100	9.718	0.428	4.162
Initial	180	9.798	0.431	4.223
Rotated 60 degrees	0	7.178	0.313	2.246
Rotated 60 degrees	100	7.203	0.316	2.279
Rotated 60 degrees	180	7.253	0.317	2.300

Table 2 Representative unprocessed records from 11.csv and 33.csv

Figure 3 uses all 201 valid records from each file without smoothing. Table 2 shows representative raw rows; the complete measurement series remains in the supplied CSV files.

4 Discussion

4.1 MPPT performance and disturbance response

The MPPT was qualitatively successful. At both the initial orientation and the 60-degree rotated position, the trajectories remained bounded, and power repeatedly returned to nearby high values after each perturbation. This is characteristic of P&O control: when the latest voltage proxy improves, the next duty-cycle step keeps its direction; when it worsens, the perturbation direction reverses. The result is a limit-cycle around the MPP rather than a perfectly constant operating point. The lower-power record also demonstrates that the same controller can continue searching after the 60-degree rotation without a new fixed duty-cycle command.

The evidence is not sufficient to claim a numerical tracking efficiency. A true efficiency requires a reference maximum power P_MPP measured by an I-V sweep at the same irradiance and cell temperature, followed by η_MPPT = P_tracked/P_MPP. The opening sample in Condition 1 is the largest recorded point, but it may be a transient and should not automatically be treated as the true MPP. The comparison is also limited because lamp irradiance and module temperature were not logged.

4.2 Buck converter and microcontroller operation

For an ideal buck converter, $V_o \approx DV_{pv}$. With a resistive load, the resistance reflected to the PV side is approximately $R_{in} = R_l/D^2$. Increasing duty cycle therefore reduces the effective resistance presented to the panel and shifts its operating voltage and current; decreasing duty cycle does the opposite. The microcontroller creates PWM, waits for the converter to settle, samples output voltage through its ADC, compares the new value with V_old, and either retains or reverses ΔD. Duty-cycle limits should be enforced in software to prevent invalid commands and excessive stress. In a general load or battery system, direct PV voltage and current measurement should replace the voltage-only proxy so that the algorithm maximises P_pv = V_pv I_pv.

4.3 Error sources and limitations

- Halogen lamps heat the module, reducing open-circuit voltage and shifting the MPP during a run.
- The perturbation step ΔD = 0.01 creates unavoidable steady-state oscillation and associated power loss.
- ADC quantisation, DACI accuracy, electrical noise, converter losses and one-second sampling affect the measured trajectory.
- The 60-degree panel rotation changes the effective irradiance, but its optical effect cannot be quantified without an irradiance sensor.

5 Improving control of temperature

Future tests should separate irradiance changes from thermal effects. The panel should be allowed to reach a defined initial temperature, and a thermocouple or RTD should be bonded to the rear surface near the centre. Temperature and irradiance should be logged with the electrical channels. A low-infrared solar simulator or LED array would reduce radiative heating; alternatively, forced airflow, a heat sink or a water-cooled rear plate could hold the module near a specified temperature. Lamp distance and tilt should be fixed with a jig, and trials should be randomised or repeated only after the panel returns to the same temperature. The reference I-V sweep and MPPT record must be obtained at matched irradiance and temperature.

6 Learning outcomes

The exercise showed that a DC-DC converter does more than change voltage: through duty-cycle control it changes the load reflected to the PV module and therefore selects the PV operating point. It also showed why a hill-climbing controller trades simplicity for a small steady-state oscillation, why settling time and perturbation size must be coordinated, and why apparently successful tracking cannot be quantified without a temperature- and irradiance-matched reference MPP. Safe sequencing, common grounding, ADC scaling and systematic data logging were as important as the algorithm itself.

7 Conclusion

The PICAXE-controlled buck converter maintained bounded operation near a high-power region at the initial orientation and after the panel was rotated by 60 degrees. Average PV power changed from 4.064 W to 2.153 W after rotation, while the controller continued its hill-climbing search. The experiment therefore met the qualitative objective of demonstrating adaptive MPPT. A future test should add panel temperature and irradiance measurements plus matched I-V sweeps so that dynamic response time and MPPT efficiency can be reported quantitatively.